SET - A

E 806

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Summative Assessment – Term III 2025 – 26

BASIC SCIENCE

Standard: VIII

Time : 2 hour
Total Score : 60

Instructions

- The first 15 minutes is cool-off time. Use this time to read and understand the questions carefully and plan your answers.
- The exam must be written in the following order: Physics, Chemistry, and Biology. Each subject is allotted 40 minutes.
- After completing each subject, the answer sheet must be handed over to the teacher.
- While writing answers, consider both score and time.
- For questions with choice, attempt only one of the given options.

PHYSICS

Time : 40 minute
Total Score : 20

Section - A

Choose the correct answer for questions 1, 2. Each question carries one score. **(2 x 1 = 2)**

1) Statement : Magnetic poles always exist in pairs; so magnets with isolated north pole do not exist.

Reason : If you break a bar magnet in half, each piece will develop its own North and South pole. (1)

a) Both Statement and Reason are true, and Reason is the correct explanation of Statement.

(b) Both Statement and Reason are true, but Reason is NOT the correct explanation of Statement.

(c) Statement is true, but Reason is false.

(d) Statement is false, but Reason is true.

2) Match the following and choose the correct option. (1)

Column A	Column B
i. Aperture	(p) The midpoint of the reflecting surface
ii. Pole	(q) Twice the focal length
iii. Radius of Curvature	(r) The diameter of the circular reflecting surface of a spherical mirror

a) i-q, ii-r, iii-p

b) i-r, ii-p, iii-q

c) 1-p, ii-r, iii-q

d) None of these

Section - B

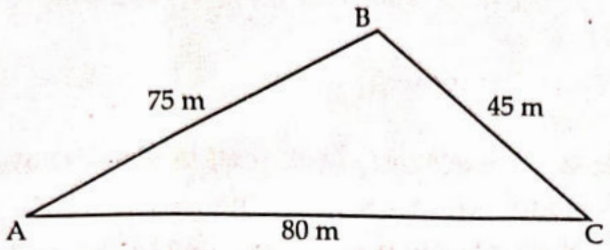
Answer questions from 3 to 8 in more than one sentence. Questions 4 and 5 have choices. Each question carries two scores.

(6 x 2 = 12)

3. If the electric charge remains in the same place without being able to flow, such electricity is called static electricity. (2)

   Write two applications of static electricity .

4. A. Observe the figure related to the path of an object.



   The object travels from A to C via B at a constant speed of 3 m/s.

   a) Calculate the time taken by the object to travel. (1)

   b) What should be the speed of the object if it travels directly from A to C in the same time? (1)

OR

4. B. A car travels from P to Q with a uniform speed of 15 m/s. The time taken for this travel is 1 hour.

   a) What do you meant by uniform speed? (1)

   b) Calculate the distance traveled by the car. (1)

5 A. Write the reason for the following

   a) The wheels of excavators are connected by wide chains. (1)

   b) Dams are built with a wider base. (1)

OR

5. B. There are situations where friction is beneficial and non beneficial to us.

   a) Write two examples of situations in everyday life where the friction is non beneficial. (1)

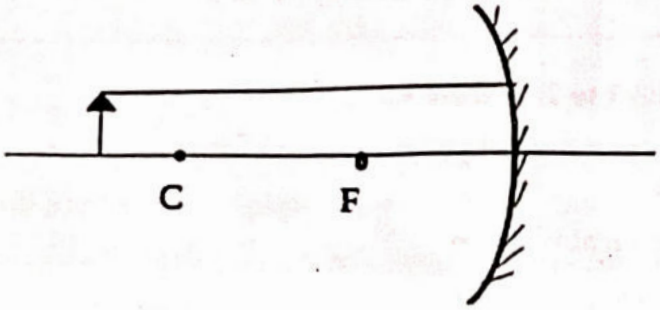
   b) Write two ways to reduce friction. (1)

6. If an object is placed between the principal focus and the pole of a concave mirror, an erect and enlarged virtual image is formed.

   a) Write down two situations in which this special feature is utilized. (1)

   b) Do we use concave mirrors as rear-view mirrors? Why? (1)

7. Observe the figure given below



Copy and complete the correct ray diagram for image formation. (2)

8. Write down the structural differences between the reflectors in torch light and street light. Write any other two situations in which the street light type reflectors are utilized. (2)

Section - C

Answer questions 9 and 10. The question 10 has a choice. Each question carries 3 score **(2 x 3 = 6)**

9. You might have heard about susceptibility and permeability in magnetism. A student has two metal rods A and B. Rod A is easy to magnetize but loses its magnetism instantly. Rod B is hard to magnetize but its magnetism stays for a long time.

   a) Identify which has got the higher susceptibility. Write the definition for that property. (2)

   b) What do you mean by permeability. (1)

10. A. The following are some statements associated with magnetism. Classify and complete the table. (3)

- Poles can be changed
- The magnetism can be retained for a long time
- Magnetic strength cannot be increased
- Poles cannot be changed
- Magnetic strength can be increased
- The magnetism cannot be retained for a long time

Electro magnets	Permanent magnets

OR

10. B. Magnetic field lines are imaginary lines used to represent the direction and strength of a magnetic field.

   a) Write down any two characteristics of magnetic field lines. (1)

   b) Copy the following diagrams. Draw the magnetic field lines associated with these situations. (2)



Figure 1



Figure 2

CHEMISTRY

Time : 40 minute
Total Score : 20

Answer all questions from 1 to 2. 1 score each. **(2 x 1 = 2)**

1. Silver chloride kept open in sunlight turns black after some time. What type of chemical reaction is this? (1)

   A) Electrochemical reaction

   B) Photochemical reaction

   C) Exothermic reaction

   D) Endothermic reaction

2. Statement (A) : When a red litmus paper is dipped in magnesium hydroxide solution, litmus paper turns blue.

   Reason (R) : Aqueous solutions of metal oxides exhibit basic nature. Which of the following is correct? (1)

   A) Both A and R are correct. R is not the correct explanation of A.

   B) A is correct. R is not correct.

   C) Both A and R are correct. R is the correct explanation of A.

   D) A is not correct. R is correct.

Two questions from 3 to 8 have choice. Each question carries 2 scores. **(6 x 2 = 12)**

3. Metals are used for various purposes in daily life. Which property of metals is utilized in the following situations? (2)

Metal	Use
Tungsten	To make filaments.
Aluminium	In the manufacture of roofing sheets

4. (A) Carbon dioxide molecule contains one carbon atom and two oxygen atoms.

   a) What is the molecular formula of carbon dioxide? (1)

   b) What is the total number of atoms in 5 carbon dioxide molecules? (1)

   OR

   (B) The molecular formula of sulphuric acid is H_2SO_4.

   a) How can three molecules of sulphuric acid be represented? (1)

   b) How many oxygen atoms are there in 5 sulphuric acid molecules? (1)

5. A child placed a blade on the surface of water in a tumbler without sinking. When the experiment was repeated using soap water, the blade sank.

   a) Which force supported the blade on the water surface? (1)

   b) Why did the experiment fail in soap water? (1)

6. Observe the picture and answer the questions.



a) What is the name of this device? (1)

b) Which are the gases produced at the positive and negative electrodes? (1)

7. Different types of glasses with various colours and properties are available.

a) What is the common component found in all types of glasses? (1)

b) Which salt is used to give blue colour to glass? (1)

8. (A) Give reasons for the following statements:

a) Clothes become cleaner when washed in soap water. (1)

b) Cooking becomes easier in pressure cookers. (1)

OR

(B). The observations of an experiment conducted using water samples from three different sources are given below. Analyse them and answer the questions.

- Sample A : Lathers well when soap is added.
- Sample B : Does not lather well when soap is added.
- Sample C : Does not lather well when soap is added, but lathers well if soap is added after heating.

a) A magnesium salt is present in sample C. Write the name of this salt? (1)

b) Write the name of any one salt that causes permanent hardness to water? (1)

One question from 9 to 10 has choice. Each question carries 3 scores. **(2 x 3 = 6)**

9. When a little Barium chloride solution is added to an acid solution taken in a test tube, a white precipitate is formed. When Hydrochloric acid was added to this precipitate, no change occurred.

a) Which is the acid solution taken here? (1)

b) What is the name of the white precipitate formed? (1)

c) Write the chemical equation of the reaction when this acid reacts with magnesium hydroxide solution. (1)

10. (A) Polymer chemistry is the branch of chemistry that deals with the study of polymers.

a) What is polymerisation? (1)

b) By what name are the small molecules involved in this reaction called? (1)

c) Which polymer is used for the manufacture of pipes? (1)

OR

(B) Plastics can be classified into two based on their nature.

a) What is the difference between Thermoplastics and Thermosetting plastics? (2)

b) How does the uncontrolled use of plastics cause soil pollution? (1)

Choices are given for questions 6, 7 and 9.

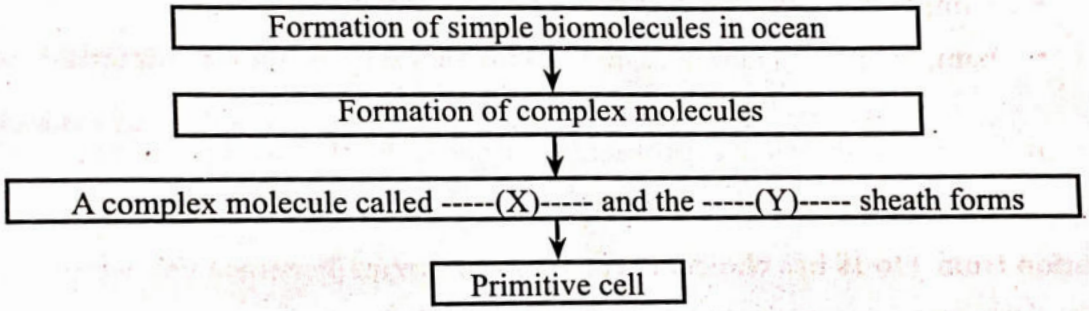
Answer the questions 1 & 2. Each question carries 1 score. **(2x1=2)**

1. Choose the correct option from the pairs formed by matching the items given in columns A and B. (1)

A. Fertilizers	B. Advantages
P. Rhizobium	i. Plants can easily absorb.
Q. Urea	ii. Contains nutrients that decomposers release from organic waste.
R. Compost	iii. Helps in the fixation of Nitrogen and Phosphorus.

a) P-iii, Q-ii, R-i
b) P-i, Q-ii, R-iii
c) P-iii, Q-i, R-ii
d) P-ii, Q-iii, R-i

2. Choose the option in which 'X' and 'Y' are correctly represented in the illustration related to the origin of life on Earth. (1)



a) X- Protein, Y- Water vapour
b) X- Nucleic acid, Y- Lipid
c) X- Methane, Y- Lipid
d) X- Oxygen, Y- Nucleic acid

Answer the questions 3 to 8. Each question carries 2 scores. **(6x2=12)**

3. Observe the figures showing the stages of cell division and answer the questions.



X Y

a) Identify the stages shown in the figures 'X' and 'Y'. (1)

b) What changes occur in 'X' and 'Y' during cell division? (1)

4. Answer the questions related to plastids.

a) What is the importance of Leucoplasts? (1)

b) How do Chromoplasts help in pollination and seed dispersal in plants? (1)

5. Analyse the statements given about twins and answer the questions.

P) The external appearance and hereditary traits will be the same.

Q) They do not necessarily have to be of the same gender.

a) Identify the type of twins mentioned in statements 'P' and 'Q'. (1)

b) How are the twins indicated in 'P' formed? (1)

6. **A.** Write two main ideas for presenting a seminar on the topic **"Is biodiversity conservation essential for the survival of human kind?"** (2)

OR

B. Write two practical suggestions to reduce the problems caused by wild animals entering human settlements. (2)

7. **A.** Analyse the given statements about a kingdom and answer the questions.

- Cell wall is made of Chitin.
- Eukaryotic saprophytes

a) Which kingdom is this? Write any one use of the organisms included in this kingdom. (1)

b) They are saprophytes. Why? (1)

OR

B. The arguments raised in a class debate are given below. Analyse them and answer the question.

Team A: Bacteria are beneficial.

Team B: Bacteria are harmful.

Give suitable evidence for both arguments. (2)

8. Evaluate the given statement and answer the questions.

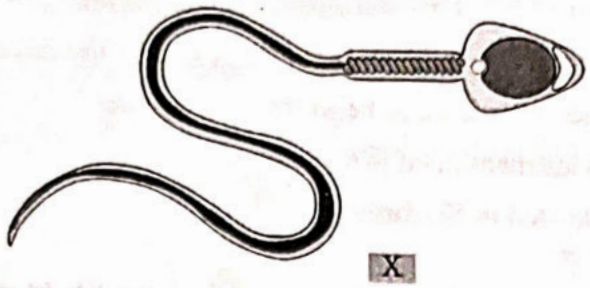
"Mutualism and commensalism are examples for interactions in nature."

a) How do the interactions mentioned in the statement differ from each other? (1)

b) Name another intereaction and give example of it. (1)

Answer the questions 9 and 10. Each question carries 3 scores. **(2x3=6)**

9. **A.** Observe the given figure and answer the questions.



X

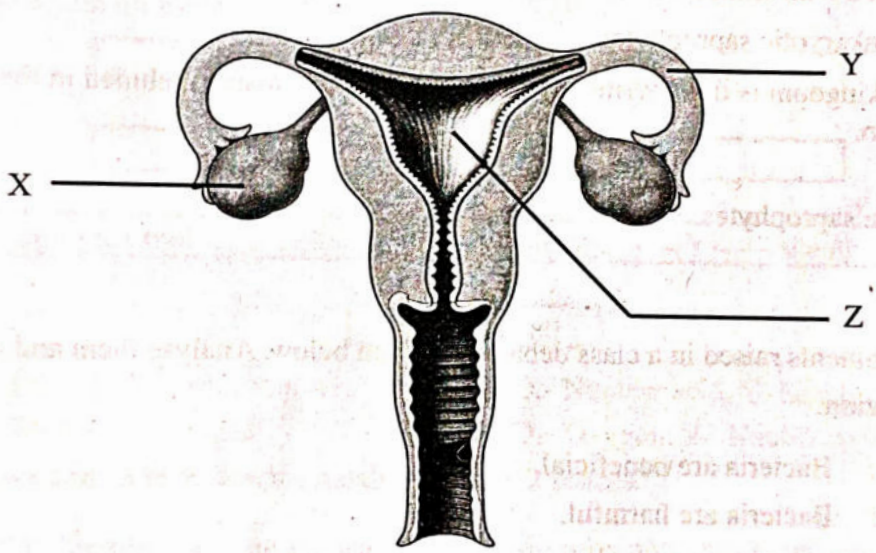
a) Identify the cell 'X'. From which part is it produced? (1)

b) Why do these cells have the ability to move? (1)

c) Is the chromosome number in these cells the same as in normal body cells? Why? (1)

OR

B. Observe the given figure and answer the questions.



a) Which hormones are produced in the part labelled as 'X'? (1)

b) How do defects in 'Y' affect reproduction? (1)

c) What is the role of the part 'Z' in reproduction? (1)

10. A food chain formed in an ecosystem is given below. Analyse it and answer the questions.

Grass → Deer → Tiger

a) Who is the secondary consumer in this food chain? Which trophic level does this organism belong to? (1)

b) Does the energy stored in Grass completely reach the tiger through this food chain? Why? (1)

c) Conservation of the tiger is necessary to maintain the balance of the ecosystem. Why? (1)